

ReactJS HOL - 13

Objective

Create a React application named **bloggerapp** using multiple components, conditional rendering, lists, keys and the `map()` function.

Theory

Various Ways of Conditional Rendering

Conditional rendering in React can be implemented using if-else statements, ternary operators, logical `&&` operators, switch statements, and by returning null.

Rendering Multiple Components

React applications can render multiple reusable components inside a parent component.

List Component

A list component displays collections of data using arrays and the `map()` function.

Keys in React

Keys uniquely identify list items and help React update the UI efficiently.

Extracting Components with Keys

Large lists can be split into reusable child components while preserving unique keys.

React `map()` Function

The `map()` function iterates through arrays and returns JSX elements for rendering.

Implementation Steps

1. Create a React project: `npx create-react-app bloggerapp`
2. Create `BookDetails`, `BlogDetails` and `CourseDetails` components.
3. Create arrays for books, blogs and courses.
4. Use `map()` to render all items.
5. Use keys for each rendered element.
6. Render the components conditionally from `App.js`.
7. Run using `npm start`.

Books.js

```
export const books=[{id:101,bname:'Master React',price:670},{id:102,bname:'Deep Dive
```

```
into Angular 11',price:800},{id:103,bname:'Mongo Essentials',price:450}]];
```

BookDetails.js

```
import {books} from './Books';  
export default function BookDetails(){return(<div><h1>Book  
Details</h1>{books.map(book=><div  
key={book.id}><h3>{book.bname}</h3><h4>{book.price}</h4></div>)}</div>);}
```

BlogDetails.js

```
export default function BlogDetails(){return(<div><h1>Blog Details</h1><h3>React  
Learning</h3><p>Welcome to learning React!</p><h3>Installation</h3><p>You can install  
React from npm.</p></div>);}
```

CourseDetails.js

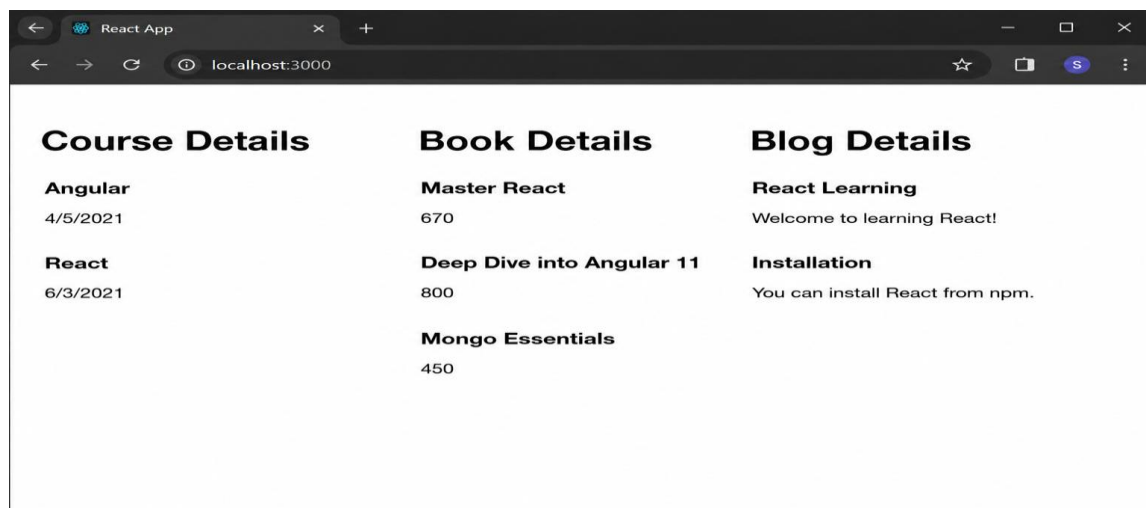
```
export default function CourseDetails(){return(<div><h1>Course  
Details</h1><h3>Angular</h3><p>4/5/2021</p><h3>React</h3><p>6/3/2021</p></div>);}
```

App.js

```
import BookDetails from './BookDetails';  
import BlogDetails from './BlogDetails';  
import CourseDetails from './CourseDetails';  
export default function App(){const flag=true;return(<div  
style={{display:'flex',gap:'40px'}}>{flag&&<CourseDetails/><BookDetails/><BlogDetails/  
></div>);}
```

Output Screenshot:

The application displays Course Details, Book Details and Blog Details in separate sections. Data is rendered using map(), each item has a unique key, and conditional rendering controls which components are displayed.



Learning Outcome

Implemented conditional rendering, multiple React components, list rendering using `map()`, and keys.

Conclusion

This exercise demonstrated reusable React components and efficient rendering of collections using the `map()` function.